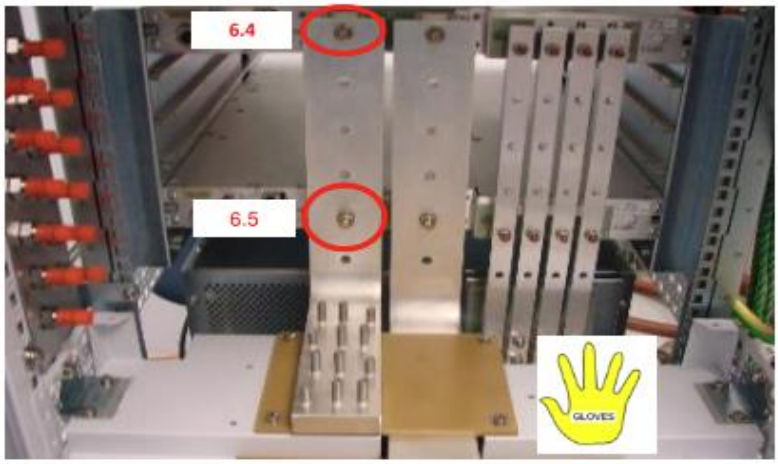
Build Overview:

1 **PSU Install**
2 **PSU AC Bus Bars Install**
3 **DC Power Shelf Install**
4 **PSU DC Bus Bars Install**



CHANGE HISTORY

Date	BPM Rev	Change Details	Updated by
	1	New Issue	JOSEPH AMAD

Prepared By:
JOSEPH R. AMAD
Manufacturing Engineer

Checked By:
IVAN NOWELL PANIQUE
Team Lead

Approved By:
SHERWIN ARIOSTE
Operations Manager

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	633-799-00	ELEC-MECH, UF2 24SLOT SUPPORT CABINET INTEGRATION	1

MACHINE: N/A

METHOD: 1. Scope of application

This BPM is suitable for UF Plus Support Cabinet Assembly
Applicable Support Cabinet Model 633-799-00
The sub-part model in Support Cabinet are 629-502-01 ADB CHASIS SUB ASSY, 653-315-00 CDBU SUB ASSY and 630-945-10 GEMINI SMC TRAY

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	640-343-00	12KW 48V PSU, 3PHASE INPUT LIQUID COOLED, FOR SEMI TEST	2
2	630-354-00	COPPER OR BRASS BUS BAR AC DIST MISCELLANEOUS	4
3	603-096-03	NUT HEX M6x10Wx5MMH W/TOOTHWSHR SS	8
4	470-501-05	WASHER FLAT M6 6.4MMID SS	8
5	472-426-12	PAN HEAD CROSS RECESSED SCREW M5X0.8 x 12MM L	41
6	638-040-00	FORGED OR STAMPED STEEL FILLER PANEL 1U MISC	4
7	479-133-18	LBL DANGER HAZ VLT 2.75X1.35"	4
8	632-070-00	PLASTIC DC SUPPLY HDW ISOLATOR PLATE	1
9	629-489-00	FORGED OR STAMPED STEEL DC DIST SHELF MISC	1
10	629-538-00	FORGED OR STAMPED STEEL ADB SHELF BKT 2U MISC	2
11	630-355-00	COPPER OR BRASS 48V RETURN DC DIST MISCELLANEOUS	1
12	629-490-00	PLASTIC DC RETURN ISOLATOR PLATE OTHER ARTICLES	1
13	472-426-20	PAN HEAD CROSS RECESSED SCREW M5X0.8 x 20MM L	4
14	634-131-00	FUSE HOLDER SINGLE POLE FUSEBLOCKS	3
15	629-481-00	PLASTIC DC SUPPLY ISOLATOR PLATER OTHER ARTICLES	1
16	470-120-12	SCRW FLAT HEAD TORX M3X.5P/12 MM SS W/NYLOK	6
17	630-356-00	COPPER OR BRASS BUS BAR 48V SUPPORT DC DIST	1
18	630-542-00	COPPER OR BRASS BUS BAR TAP DC DIST MISCELLANEOUS	3
19	603-096-01	NUT HEX SS M8X1.25 WSGR W/TTH 13MM WDH	4
20	470-501-06	WASHER FLAT 8.4MM ID 16MM OD SS	6
21	523-013-03	FUSE KLK-D 3AMP FAST ACTING	3
22	614-766-90	FASTNR RP-017-1 12X30X10MM HT. M8 THRD, HOLD ZN. ALLOY	2
23	470-933-01	SCREW CAO SKT BUTHD M8x1.25 20MML SS	2
24	470-502-07	WASHER LK SPLT M8 14.8MM OD SS	2
25	634-678-00	CABLE 48VDC(-) TO GROUND W/ CONN	1
26	479-133-18	LBL DANGER HAZ VLT 2.75X1.35"	1
27	634-999-00	FORGED OR STAMPED STEEL ACCESS COVER DC DIST SHELF	1
28	603-096-00	NUT, HEX, SS M8X1.25, WSHR W/TTH, 13MM WDH	2

MACHINE:

NO.	DESCRIPTION	PARAMETER
1	10mm Socket Head	5.6 N.m
2	PH2 Bit	3.3 N.m
3	8mm Socket Head	3.3 N.m
4	T9 Torx Bit	0.7 N.m
5	Flat Head Screwdriver Bit	0.7 N.m
6	13mm Socket Gead	13.5 N.m
7	5mm Hexagon Head	13.5 N.m

MACHINE:

NO.	DESCRIPTION	PARAMETER
1	10mm Socket Head	5.6 N.m
2	PH2 Bit	3.3 N.m
3	8mm Socket Head	3.3 N.m
4	T9 Torx Bit	0.7 N.m
5	Flat Head Screwdriver Bit	0.7 N.m
6	13mm Socket Gead	13.5 N.m
7	5mm Hexagon Head	13.5 N.m

METHOD:

Flow chart of operation steps

BPM preparation	See (1. Scope of application)
↓	
Preparation Tools	See (2. Tools/Materials/Parameters---1. Tools)
↓	
Prepare Materials	See (2. Tools/Materials/Parameters---2. Materials)
↓	
Complete Operation	See (3. Operation Steps)
↓	
Inventory	See (2. Tools/Materials/Parameters---2. Materials)
↓	
Operation Check	See (4. Matters needing attention)
↓	
Inventory collection tools	See (2. Tools/Materials/Parameters---1. Tools)

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	640-343-00	12KW 48V PSU, 3PHASE INPUT LIQUID COOLED, FOR SEMI TEST	2

MACHINE: N/A

METHOD:

1.1 Remove the outer packaging of the 2 PSUs (640-343-00) and check the incoming materials.

a. Whether the power stud is bent or damaged.

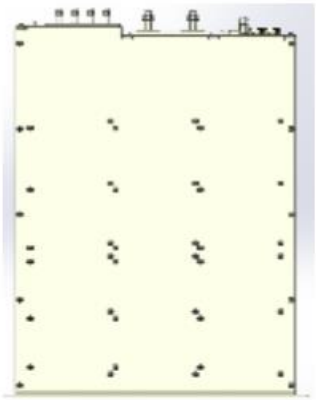
b. Whether the quick disconnect connector is bent or damaged.

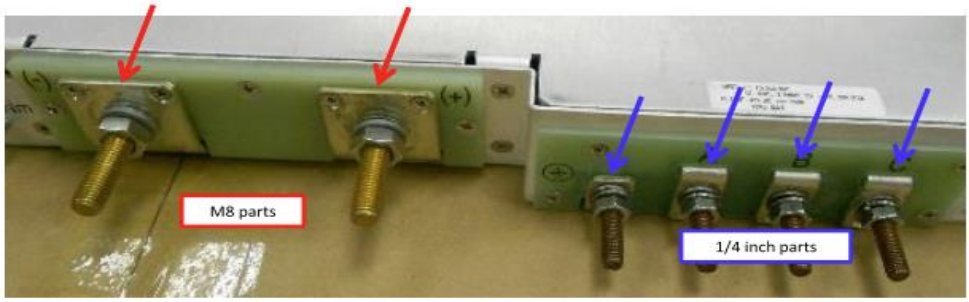
c. Whether the PSU outer packing box is damaged by impact.

1.2 Remove the nuts, washers and other parts on the Busbar of the PSU and save them for subsequent assembly.

a. 1/4 inch parts on AC connector-4 sets of nuts, flat washers and spring washers. The removal tool is a 10mm socket wrench.

b. M8 parts on the DC connector-2 sets of nuts, flat washers and spring washers. The removal tool is a 13mm socket wrench.





MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

1.3 Install the PSU into the power supply as shown in the figure, the positions are 0 (the highest position) and 4 (the second lowest position) slide into the rail until the back of the PSU is flush with the back of the ADB.

Note: The PSU is heavy and two people are required for installation.

Cabinet maintenance surface

Cabinet operation surface

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	630-354-00	COPPER OR BRASS BUS BAR AC DIST MISCELLANEOUS	4
2	603-096-03	NUT HEX M6x10Wx5MMH W/TOOTHWSHR SS	8
3	470-501-05	WASHER FLAT M6 6.4MMID SS	8

MACHINE:

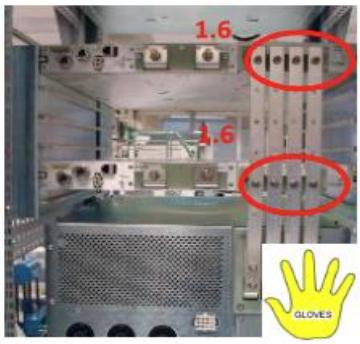
NO.	DESCRIPTION	PARAMETER
1	10mm Socket Head	4.5 N.m

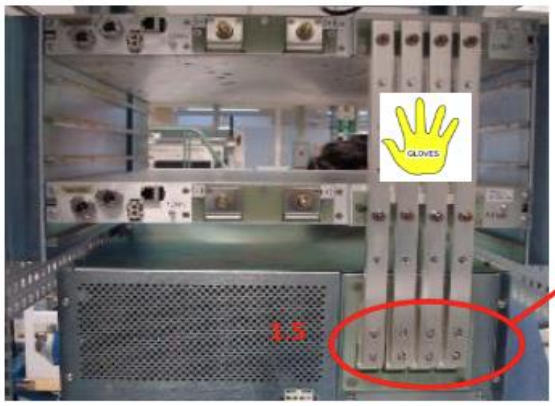
METHOD:

1.4 As shown in the figure, install 4 AC Busbars (630-354-00) on the ADB and 2 PSUs.

1.5 First fix ADB as shown in the figure, use 8 M6 lock nuts (603-096-03) and 8 M6 flat washers (470-501-05) to fix two Busbar positions. If necessary, move the position of the PSU and ADB to ensure that the Busbar is flush with the ADB and PSU. Use tool 10mm socket wrench, torque value is 5.6 N·m in 8 positions

Note: After the Busbar is flush with the ADB, adjust the position of the PSU, and the PSU can be moved back during operation. Just make sure that the PSU studs are aligned with the Busbar hole.





MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE:

ITEM	DESCRIPTION	PARAMETER
1	PH2 BIT	3.3 N.m

METHOD:

1.7 At this time, fix the ADB on the cabinet rail as shown in the figure. The tool is a PH2 Phillips torque screwdriver with a torque value of 3.3 N·m.

ADB

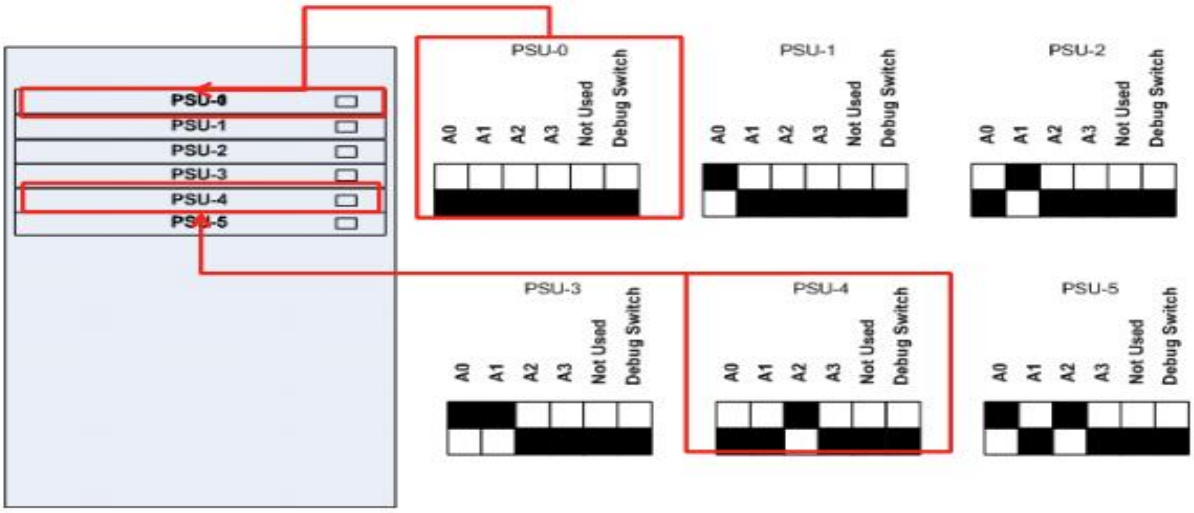
MAN: CERTIFIED OPERATOR, TECHNICIAN

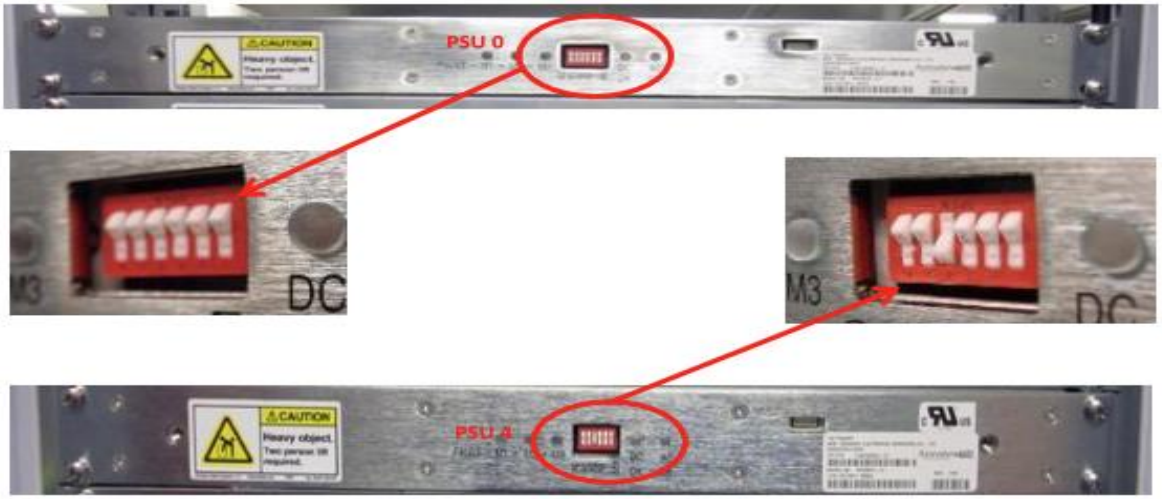
MATERIAL: N/A

MACHINE: N/A

METHOD:

1.8 Set the DIP switch on the front of the PSU on the maintenance surface of the cabinet as shown in the figure



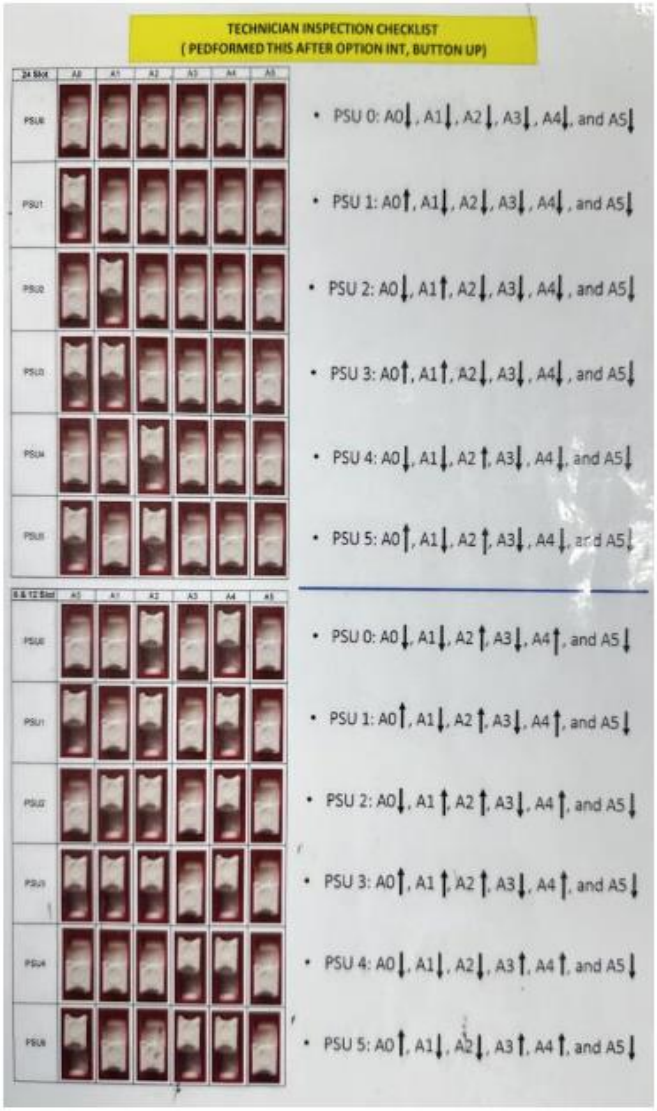


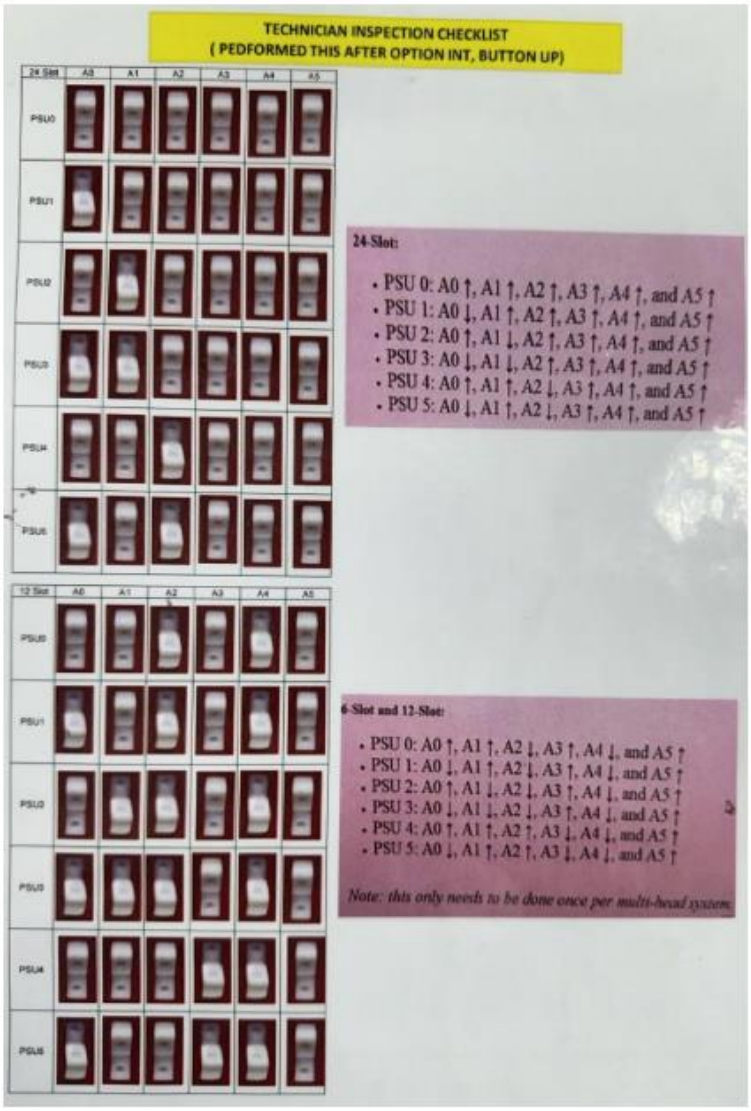
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:





MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	472-426-12	PAN HEAD CROSS RECESSED SCREW M5X0.8 X 12MM L	8

MACHINE:

ITEM	DESCRIPTION	PARAMETER
1	PH2 BIT	3.3 N.m

METHOD:

1.9 Use 8 M5 screws (472-426-12) to attach the PSU to the bracket as shown in the picture.
The tool is a PH2 Phillips torque screwdriver with a torque value of 3.3 N · m.

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	638-040-00	FORGED OR STAMPED STEEL FILLER PANEL 1U MISC	4
2	472-426-12	PAN HEAD CROSS RECESSED SCREW M5X0.8 X 12 MM L	16
3	479-133-18	LBL DANGER HAZ VLT 2.75X1.35"	4

MACHINE:

NO.	DESCRIPTION	PARAMETER
1	PH2 BIT	3.3 N.m
2	8MM SOCKET HEAD	3.3 N.m

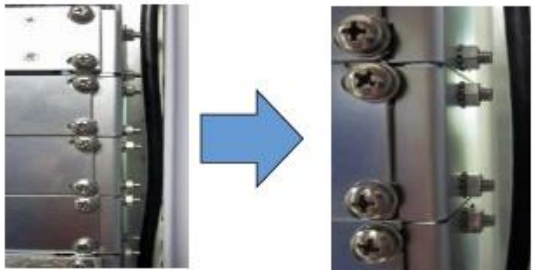
METHOD:

1.10 As shown in the figure, install 4 PSU baffles (638-040-00) on the remaining positions of PSU 1, 2, 3 and 5. Use 16 M5 screws (472-426-12) to fix, the tool is a PH2 Phillips torque screwdriver, the torque value is 3.3 N·m.

Note: 1. When installing the baffle, make sure that the part number on the baffle faces inward.
2. Ensure that all plates are flat at same level.

1.11 Adjust the position of the baffle on the bracket as shown in the figure to ensure that there is the same distance between each other.

1.12 Tighten the nut on the bracket as shown in the figure. The tool is an 8mm open-end wrench and the torque value is 3.3 N·m.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	632-070-00	PLASTIC DC SUPPLY HDW ISOLATOR PLATE	1
2	629-489-00	FORGED OR STAMPED STEEL DC DIST SHELF MISC	1
3	472-426-12	PAN HEAD CROSS RECESSED SCREW M5X0.8 x 12MM L	6
4	629-538-00	FORGED OR STAMPED STEEL ADB SHELF BKT 2U MISC	2
5	630-355-00	COPPER OR BRASS 48V RETURN DC DIST MISCELLANEOUS	1
6	629-490-00	PLASTIC DC RETURN ISOLATOR PLATE OTHER ARTICLES	1
7	472-426-20	PAN HEAD CROSS RECESSED SCREW M5X0.8 x 20MM L	2

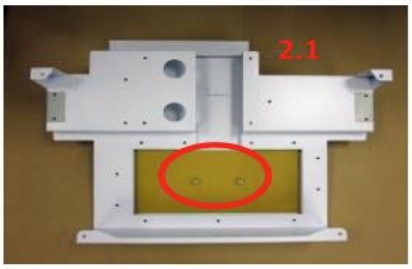
MACHINE:

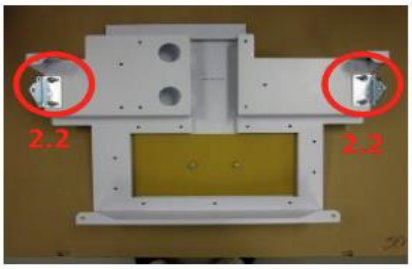
ITEM	DESCRIPTION	PARAMETER
1	PH2 BIT	3.3 N.m

METHOD:

2.1 As shown in the figure, install an insulating board (632-070-00) on a power supply rack (629-489-00). Use 2 M5 screws (472-426-12) to fix, the tool is PH2 Phillips torque screwdriver, the torque value is 3.3 N·m.

2.2 Install the two brackets (629-538-00) on both sides of the power supply rack as shown in the figure, and fix them with 4 M5 screws (472-426- 12). Do not tighten them at this time and keep them slack.





2.3 As shown in the figure, install a Busbar (630-355-00) on an insulating board (629-490-00), use 2 M5 screws (472-426-20) to fix, the tool is PH2 cross torque Screwdriver with a torque value of 3.3 N·m.

Note: The corner direction of the red circle on the insulation board is placed as shown in the figure.

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	634-131-00	FUSE HOLDER SINGLE POLE FUSEBLOCKS	3
2	629-481-00	PLASTIC DC SUPPLY ISOLATOR PLATE OTHER ARTICLES	1
3	470-420-12	SCRW FLAT HEAD TORX M3X.5P/12 MM SS W/NYLOK	6
4	630-356-00	COPPER OR BRASS BUS BAR 48V SUPPLY DC DIST	1
5	472-426-20	PAN HEAD CROSS RECESSED SCREW M5X0.8 x 20MM L	2

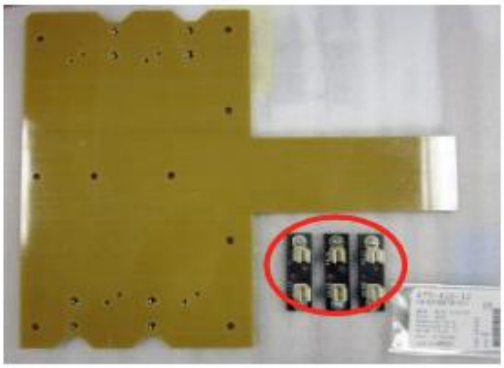
MACHINE:

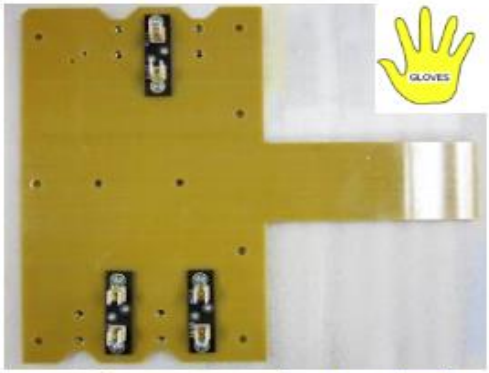
ITEM	DESCRIPTION	PARAMETER
1	T9 TORX BIT	0.7 N.m
2	PH2 BIT	3.3 N.m

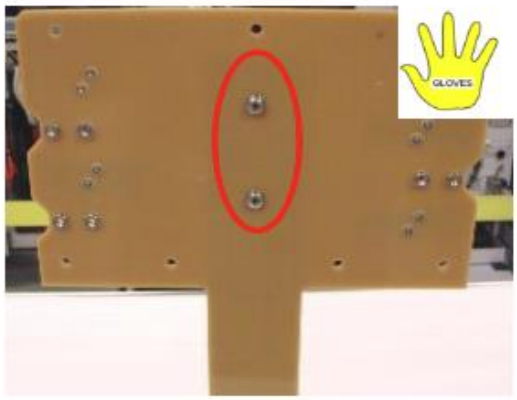
METHOD:

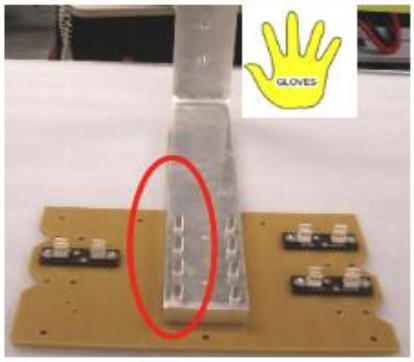
3.1 Install the 3 fuse holders (634-131-00) on the 48V insulation board (629-481-00) as shown in the figure, use 6 M3 Torx screws (470-420-12), and the tool is T9 Torx The torque wrench has a torque value of 0.7 N·m.

3.2 Install 1 Busbar (630-356-00) on the insulating board as shown in the figure, and fix it with 2 M5 screws (472-426-20), the tool is a PH2 Phillips torque screwdriver, and the torque value is 3.3 N·m . The screw installation direction is behind the insulating board.









MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	630-542-00	COPPER OR BRASS BUS BAR TAP DC DIST MISCELLANEOUS	3
2	603-096-01	NUT HEX SS M8X1.25 WSHR W/TTH 13MM WDH	3
3	470-501-06	WASHER FLAT 8.4MM ID 16MM OD SS	3
4	523-013-03	FUSE KLK-D 3AMP FAST ACTING	3

MACHINE:

ITEM	DESCRIPTION	PARAMETER
1	FLAT HEAD SCREWDRIVER BIT	0.7 N.m
2	13MM SOCKET HEAD	13.5 N.m

METHOD:

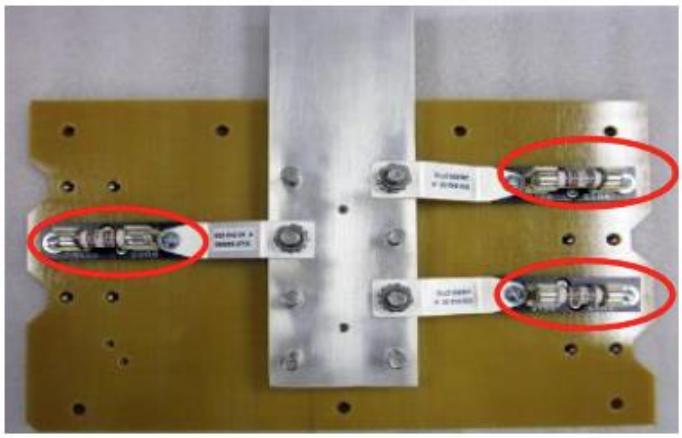
3.3 Install 3 small Busbars (630-542-00) between the fuse and the middle main Busbar as shown in the figure.

a. Fix the Busbar on the fuse holder, use the existing screws, the tool is a flat-blade screwdriver, and the torque value is 0.7 N·m.

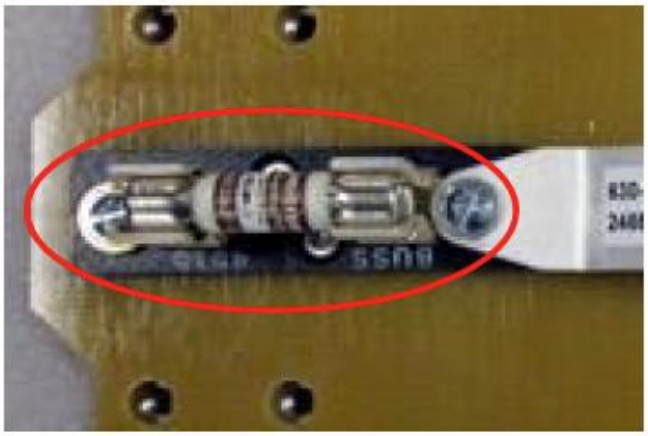
b. Fix the Busbar on the middle main Busbar, use 3 M8 hexagon nuts (603-096-01), 3 M8 flat washers (470-501-06), use a tool 13mm socket wrench, torque value 13.5 N ·M.

3.4 Install the 3 fuses (523-013-03) on the fuse holder as shown in the figure, and then press the two ends of the fuse to fully snap into the holder.

3.3



3.4



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

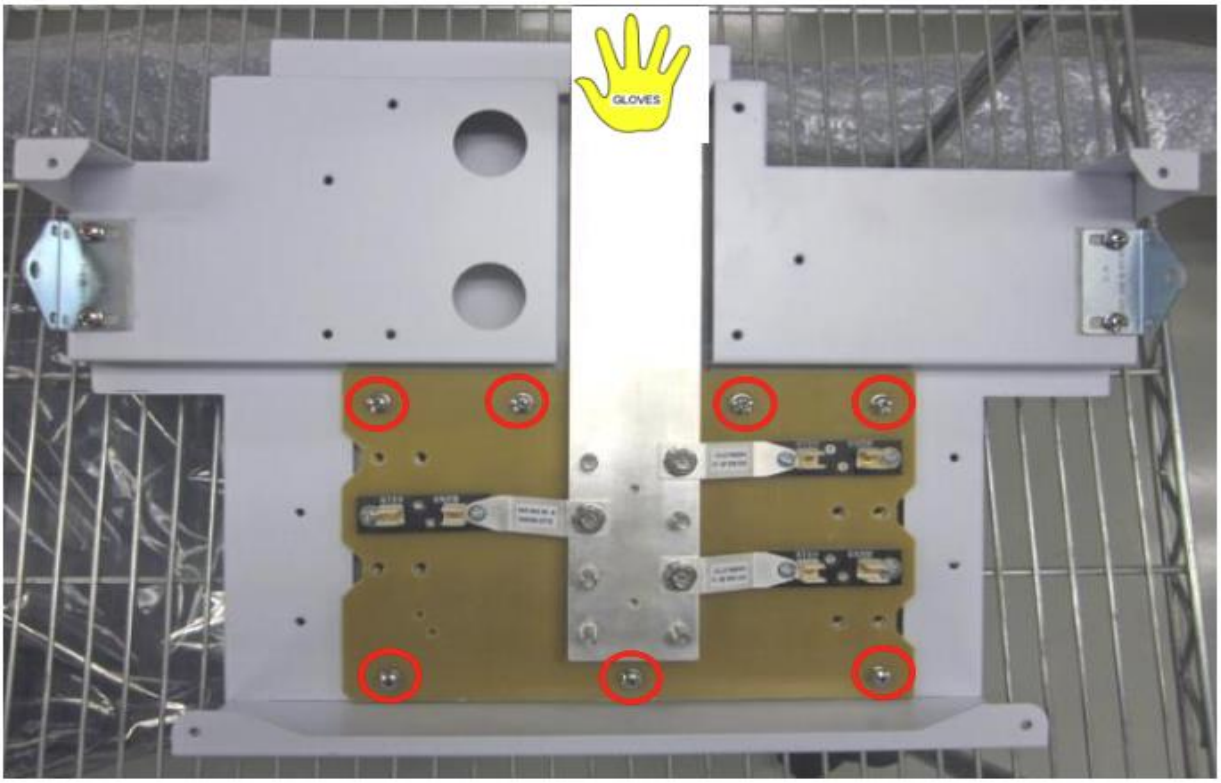
ITEM	P/N	DESCRIPTION	QTY
1	472-426-12	PAN HEAD CROSS RECESSED SCREW M5X0.8 X 12MM L	7

MACHINE:

ITEM	DESCRIPTION	PARAMETER
1	PH2 BIT	3.3 N.m

METHOD:

5.1 Install the assembled 48V Busbar to the power supply bracket as shown in the figure, and fix it with 7 M5 screws (472-426-12), the tool is a PH2 Phillips torque screwdriver, and the torque value is 3.3 N·m.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

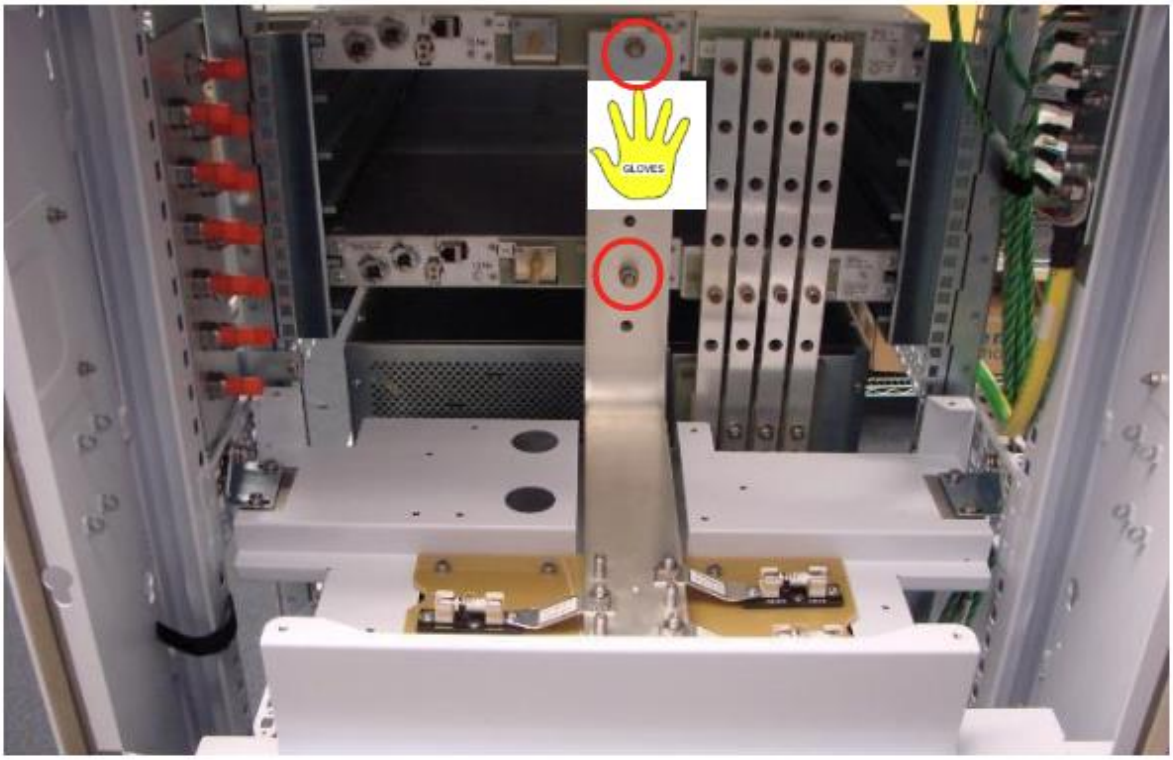
MACHINE: N/A

METHOD:

5.2 Ensure that the brackets on both sides of the power supply rack are loose.

5.3 Carefully take the power supply rack, which has the previously installed 48V Busbar on it, and place them on the operating surface of the cabinet as shown in the figure. Align the 48V Busbar with the output port on the PSU.

Install the 8mm nuts, flat washers and spring washers that were removed from the PSU before, and manually tighten the nuts.



MAN: CERTIFIED OPERATOR, TECHNICIAN

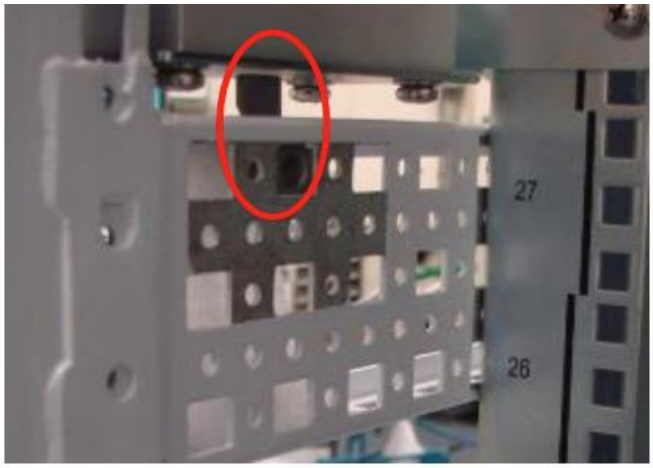
MATERIAL:

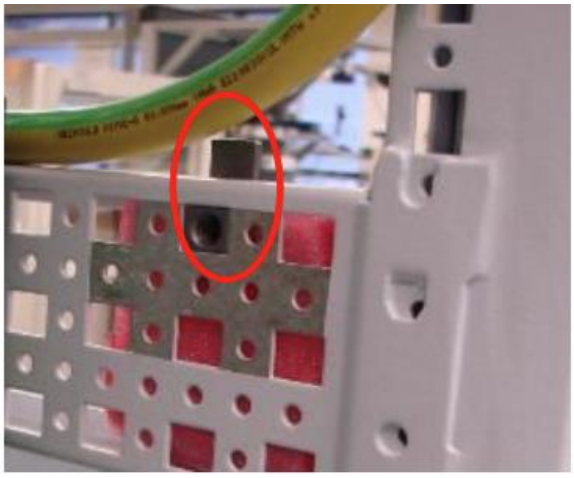
ITEM	P/N	DESCRIPTION	QTY
1	614-766-90	FASTNR RP-017-1 12X30X10MM HT. M8 THRD. HOLE ZN.ALLOY	2
2	470-993-01	SCREW CAP SKT BUTHD M8X1.25 20MML SS	2
3	470-502-07	WASHER LK SPLT M8 14.MM OD SS	2
4	470-501-06	WASHER FLAT 8.4MM ID 16MM OD SS	2

MACHINE: N/A

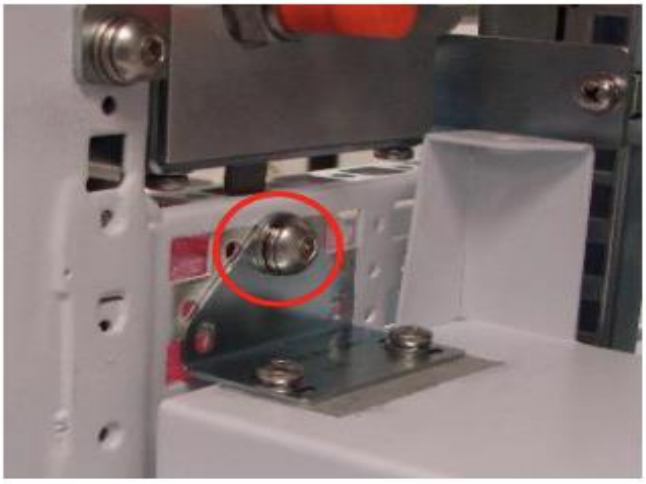
METHOD:

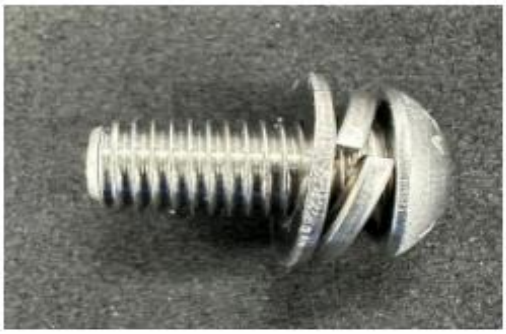
5.4 Install two clamps (614-766-90) on the side rails of the cabinet as shown in the figure.





5.5 Install an M8 Spring Washer (470-502-07) and an M8 Flat Washer (470-501-06) on the M8 Screw (470-933-01) as shown.





MAN: CERTIFIED OPERATOR, TECHNICIAN

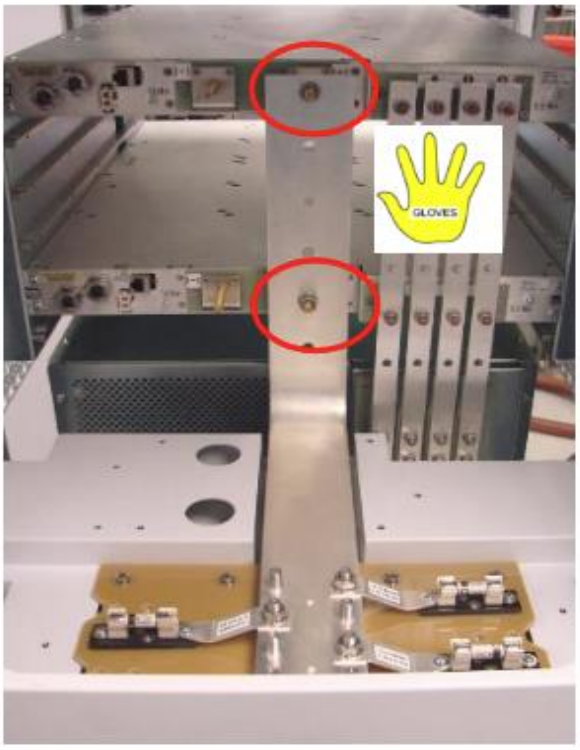
MATERIAL: N/A

MACHINE:

ITEM	DESCRIPTION	PARAMETER
1	13MM SOCKET HEAD	10 N.m

METHOD:

5.6 Finally, fix the Busbar nut on the PSU, the tool is a 13mm socket wrench, and the torque value is 10 N·m.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	472-426-12	PAN HEAD CROSS RECESSED SCREW M5X0.8 X 12MM L	4

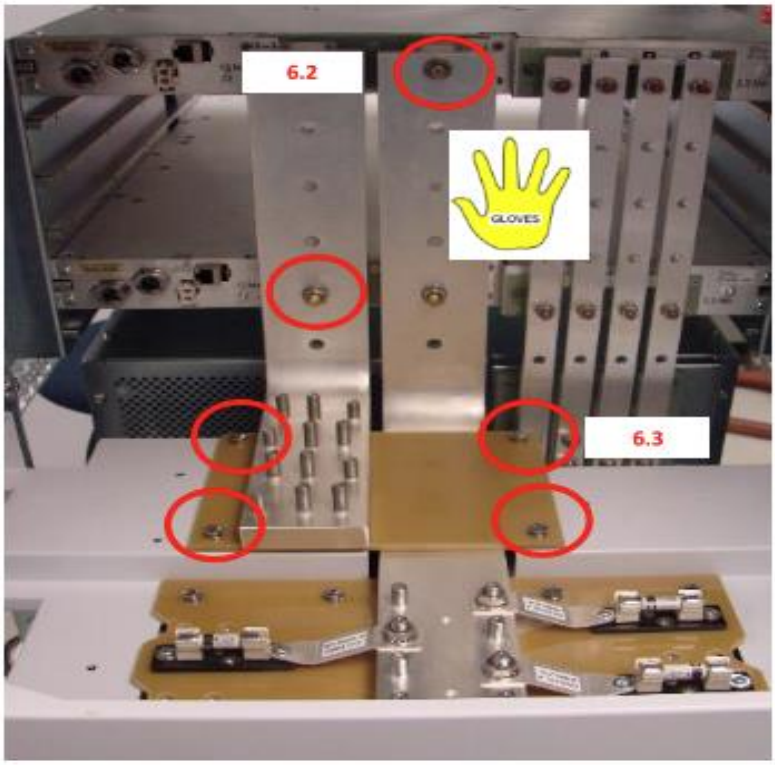
MACHINE: N/A

METHOD:

6.1 Carefully pick up the assembled 48V Busbar as shown in the figure and put it into the operating surface of the cabinet. Align the 48V Busbar with the return port on the PSU.

6.2 Install the 8mm nuts, flat washers and spring washers removed from the PSU as shown in the picture and tighten manually.

6.3 Fix the Busbar on the power supply rack as shown in the figure, use 4 M5 screws (472-426-12) to fix, and tighten the screws



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE:

ITEM	DESCRIPTION	PARAMETER
1	13MM SOCKET HEAD	10 N.m
2	PH2 BIT	3.3 N.m
3	5MM HEXAGON HEAD	13.5 N.m

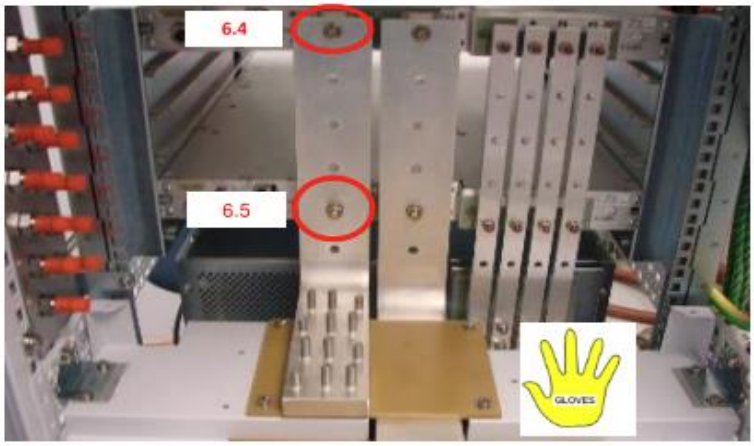
METHOD:

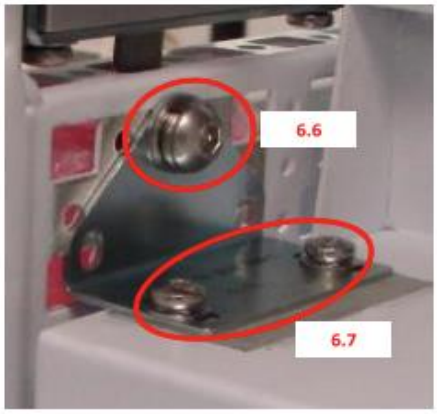
6.4 Use 2 nuts to fix the Busbar on the PSU as shown in the figure. The tool is a 13mm socket wrench and the torque value is 10 N·m.

6.5 Fix the 4 insulation board screws as shown in the figure. The tool is a PH2 Phillips torque screwdriver, and the torque value is 3.3 N·m.

6.6 As shown in the figure, use 2 M8 nuts to fix the bracket on the side guide rail. The tool is a 5mm hexagonal bit with a torque value of 13.5 N·m.

6.7 As shown in the figure, use 4 M5 nuts to fix the bracket on the power rack. The tool is a PH2 Phillips torque screwdriver with a torque value of 3.3 N·m.





MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

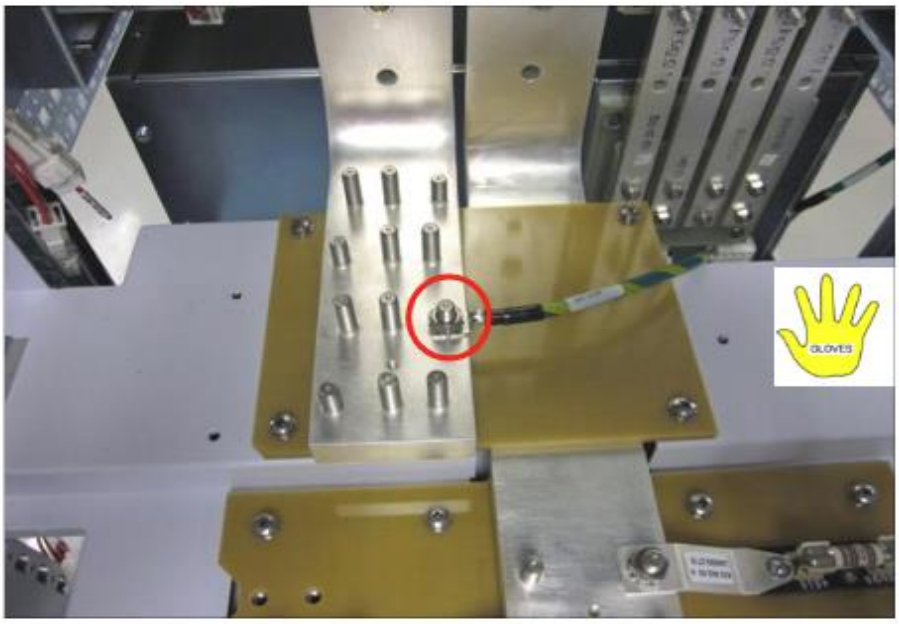
ITEM	P/N	DESCRIPTION	QTY
1	634-678-00	CABLE 48 VDC(-) TO GROUND W/CONN	1
2	470-501-06	WASHER FLAT 8.4MM ID 16MM OD SS	1
3	603-096-01	NUT HEX SS M8X1.25 WSHR W/TTH 13MM WDH	1

MACHINE:

NO.	DESCRIPTION	PARAMETER
1	13MM SOCKET HEAD	13.5 N.m

METHOD:

6.7 Connect the other end (634-678-00) of the grounding panel MECCA G5 ground wire to the 48V Busbar as shown in the figure, using 1 M8 flat washer (470-501-06) and 1 M8 hexagon nut (603-096-01) Fixed, the tool is a 13mm socket wrench, and the torque value is 13.5 N·m.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	479-133-18	LBL DANGER HAZ VLT 2.75X1.35"	1
2	634-999-00	FORGED OR STAMPED STEEL ACCESS COVER DC DIST SHELF	1
3	603-096-00	NUT HEX SS M8X1.25 WSHR W/TTH 13MM WDH	2

MACHINE:

NO.	DESCRIPTION	PARAMETER
1	8MM SOCKET HEAD	3.3 N.m

METHOD:

7.1 Install a warning label (479-133-18) on the AC cover (634-999-00) as shown in the figure, and place the label on the center of the panel.

7.2 Install the AC cover plate (634-999-00) on the cabinet as shown in the figure, and fix it with two M5 nuts (603-096-00), the tool is an 8mm socket wrench, and the torque value is 3.3 N·m

Note: Ensure that the ground wire is behind the guide rail and in the opening slot of the cover.